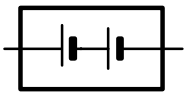
GLOSSARY OF TERMS AND SYMBOLS

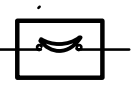


BATTERY

Stores chemical energy and converts it into electrical energy. Provides DC current for the auto's various electrical circuits.

CAPACITOR (Condenser)

A small holding unit for temporary storage of electrical voltage.



CIGARETTE LIGHTER

An electric resistance heating element.

CIRCUIT BREAKER

Basically a reusable fuse, a circuit breaker will heat and open if too much current flows through it. Some units automatically reset when cool, others must be manually reset.

DIODE

A semiconductor which allows current flow in only one direction.

DIODE, ZENER

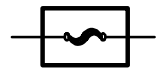
A diode which allows current flow in one direction but blocks reverse flow only up to a specific voltage. Above that potential, it passes the excess voltage. This acts as a simple voltage regulator.

DISTRIBUTOR, IIA

Channels high–voltage current from the ignition coil to the individual spark plugs.

FUSE

A thin metal strip which burns through when too much current flows through it, thereby stopping current flow and protecting a circuit from damage.



(for Medium Current Fuse)

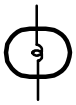
FUSIBLE LINK

A heavy–gauge wire placed in high amperage circuits which burns through on overloads, thereby protecting the circuit. The numbers indicate the cross–section surface area of the wires.

(for High Current Fuse or Fusible Link.)

GROUND

The point at which wiring attaches to the Body, thereby providing a return path for an electrical circuit; without a ground, current cannot flow.



HEADLIGHTS

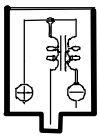
1. SINGLE FILAMENT

Current flow causes a headlight filament to heat up and emit light. A headlight may have either a single (1) filament or a double (2) filament.

2. DOUBLE FILAMENT

HORN

An electric device which sounds a loud audible signal.



IGNITION COIL

Converts low–voltage DC current into high–voltage ignition current for firing the spark plugs.

LIGHT

Current flow through a filament causes the filament to heat up and emit light.

LED (LIGHT EMITTING DIODE)

Upon current flow, these diodes emit light without producing the heat of a comparable light.

METER, ANALOG

Current flow activates a magnetic coil which causes a needle to move, thereby providing a relative display against a background calibration.

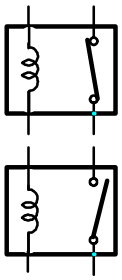
METER, DIGITAL

Current flow activates one or many LED's, LCD's, or fluorescent displays, which provide a relative or digital display.

MOTOR

A power unit which converts electrical energy into mechanical energy, especially rotary motion.

RELAY

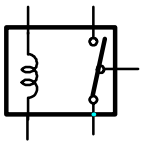


1. NORMALLY CLOSED

2. NORMALLY OPEN

Basically, an electrically operated switch which may be normally closed (1) or open (2).

Current flow through a small coil creates a magnetic field which either opens or closes an attached switch.



RELAY, DOUBLE THROW

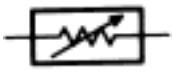
A relay which passes current through one set of contacts or the other.

RESISTOR

An electrical component with a fixed resistance, placed in a circuit to reduce voltage to a specific value.

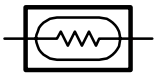
RESISTOR, TAPPED

A resistor which supplies two or more different non adjustable resistance values.



RESISTOR, VARIABLE or RHEOSTAT

A controllable resistor with a variable rate of resistance. Also called a potentiometer or rheostat.



SENSOR (Thermistor)

A resistor which varies its resistance with temperature.

SENSOR, ANALOG SPEED

Uses magnetic impulses to open and close a switch to create a signal for activation of other components.

SHORT PIN

Used to provide an unbroken connection within a junction block.

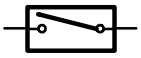
SOLENOID

An electromagnetic coil which forms a magnetic field when current flows, to move a plunger, etc.

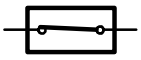
SPEAKER

An electromechanical device which creates sound waves from current flow.

SWITCH, MANUAL



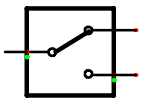
1. NORMALLY OPEN



2. NORMALLY CLOSED

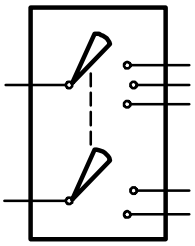
Opens and closes circuits, thereby stopping (1) or allowing (2) current flow.

SWITCH, DOUBLE THROW



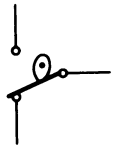
A switch which continuously passes current through one set of contacts or the other.

SWITCH, IGNITION



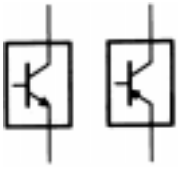
A key operated switch with several positions which allows various circuits, particularly the primary ignition circuit, to become operational.

SWITCH, WIPER PARK



Automatically returns wipers to the stop position when the wiper switch is turned off.

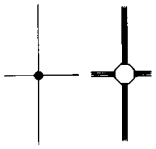
TRANSISTOR



A solidstate device typically used as an electronic relay; stops or passes current depending on the voltage applied at “base.”

WIRES

(1) NOT CONNECTED



(2) SPLICED

Wires are always drawn as straight lines on wiring diagrams. Crossed wires (1) without a black dot at the junction are not joined; crossed wires (2) with a black dot or octagonal (◯) mark at the junction are spliced (joined) connections.